

(20M)

Medium Access Control Layer

23/3/22

⇒ MAC: It is a sublayer of DLL, its main job or objective is to control how different devices use the communication channel

⇒ Why MAC layer is needed? In a network many devices are connected to the same medium (wifi or lan) if all devices try to send data at same time, the signals will overlap & collide so, Bcz of this data may be lost. so we need MAC layer to control this situation. It makes sure that the devices don't transmit at same time unnecessarily. It ensures that devices transmit data in a controlled and sequential manner

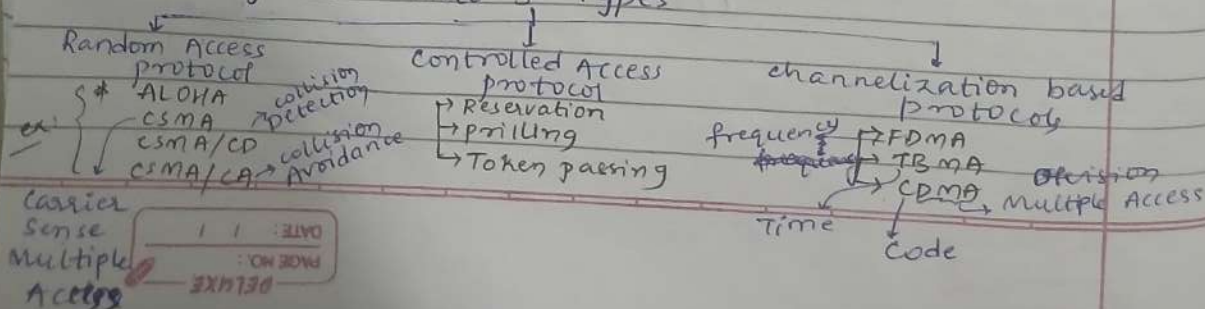
⇒ Types of Communication

MAC layer supports different modes of communication based on how data is delivered between devices in a network

- 1) Unicast: In this type of Comm, data is transmitted from one device to the specific destination device. It is most common form of comm in network. Also known as one-to-one comm
- 2) Multicast: In this type, data is transmitted from one device to a selected group of devices. Only the devices that are part of the group receive data. Also known as one-to-many comm
- 3) Broadcast: Data is transmitted from one device to all devices present in the network. In this type, every device connected to the network, receives the data (one-to-all comm)

⇒ The MAC layer provides Full duplex communication, which means that data can be sent & received simultaneously eg: Telephone

⇒ Types of MAC protocols: MA protocols are used when many devices share same comm channel, These protocols decide how and when a device can send data to avoid collision and ensure the smooth communication. MA protocols are categorized into 3 types



Random Access Protocols are used in networks where multiple devices share same communication channel. In this method any device can transmit data, whenever it has data to send and here devices do not follow any fixed order while sending data. So, Bcz this freedom sometimes two or more devices may transmit at the same time which leads to collision. To handle this these protocols use techniques to detect, avoid or recover from collisions. After a collision occurs devices wait for a random period of time before re-transmitting the data. These protocols are suitable for networks where data transmission is irregular. Eg: ALOHA is random access protocol used in CN where multiple devices share same comm channel. Main Concept of ALOHA is a device can transmit data whenever it has data to send without checking whether the channel is free or busy. Here, there is no proper control, two or more ^{devices} may send data at the same time. So, Bcz of this collision occurs and the data is not received properly in such cases the sender understands that the transmission has failed & sends the data again. After sending a frame the sender waits for Acknowledgment, if Ack is received it means frame is successful. If the Ack does not come within a certain time, the sender assumes that the data was not received, mostly likely due to collision. It then waits for some random time (& this is also called Backoff time) and retransmits the frame. This random delay reduces the chances of collisions. It is of 2 types

^{ALOHA}
 Pure ALOHA Slotted ALOHA

⇒ PURE ALOHA: In this method, device transmits data immediately whenever it has data. There is no synchronization, so, collisions can occur at any time. Bcz of this performance is low. Vulnerable time (V_t) is the time period, during which a transmitted frame has risk of collision. Calculated by $[V_t = 2 * T_f]$. Here T_f is the total required by a device to send one frame over channel.

$$T_f = \frac{\text{frame size}}{\text{Transmission rate (bps)}}$$

$$V_t = \frac{2 \times \text{frame size}}{\text{Transmission rate (bps)}}$$

→ $S = G \cdot e^{-2G}$ ← for PURE ALOHA

throughput represents the no. of frames that are successfully transmitting over the network

→ $S = \text{Throughput}$

$G = \text{Total no. of transmission attempts including re-transmissions}$

→ for max throughput $\frac{dS}{dG} = 0$

→ $G \cdot e^{-2G} (-2) + e^{-2G} = 0$

→ $e^{-2G} (-2G + 1) = 0$

→ $-2G + 1 = 0$

→ $G = 1/2$

∴ $S = 1/2 \cdot e^{-1/2} \Rightarrow S = 0.184$

→ $S = 18.4\%$

9) A pure ALOHA system transfers frames of 1200 bits over a channel with a bandwidth 1.5 Mbps. It is given that on avg. time there are 0.8 transmission attempts per frame time. Calculate V_f & S (throughput)

$B = 1.5 \text{ Mbps}$, $T_f = \frac{\text{frame size}}{B(\text{bps})} = \frac{1200}{1.5 \times 10^6} = 0.8 \times 10^{-3} = 0.8 \text{ ms}$

$V_f = 2 \times T_f = 1.6 \text{ ms}$

$S = G \cdot e^{-2G} = 0.8 \times e^{-2 \times 0.8} = 0.8 \times e^{-1.6} = 0.16 \Rightarrow 16\%$

→ 16% are transmitted successfully

⇒ Slotted

: In this time is divided into equal intervals (time slots). Each slot is equal to time required to transmit one frame. In this approach a device is allowed to transmit data only at beginning of a time slot not at any random time and this is key diff. from PURE ALOHA

→ Because of this synchronization, the chances of collision are reduced. However, collisions still occur, if 2 or more devices transmit in the same time slot

Working

- 1) The channel
- 2) Each device
- 3) If only 2 d transmission in same
- 4) After collision and then re-Throughput:

This means one time slot

⇒ max through

9) A N/w admin and slotted ALO an avg of 1.2 transmission time for both cases

i) $G = 1.2$, pure slot

ii) % improvement

CSMA: a de already used MA → Multiple a same com

CSMA: Is a pr channel and eac to make sure i

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bits over a channel

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ulate V_t & S (throughput)

$0.8 \times 10^{-3} = 0.8 \text{ ms}$

$1.6 = 0.16 \Rightarrow 16\%$
 $G = 50\%$ ↑ less

d into equal intervals
required to transmit
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time slot not at any

PURE ALOHA

any of collision reduced

u, if 2 or more
same time slot

Working

- 1) The channel time is divided into fixed time slots
- 2) Each device waits for the start of the next slot before transmitting
- 3) If only 1 device transmits in its slot, the transmission is successful, If more than 1 device transmits in same slot collision occurs
- 4) After collision the sender waits for a random back-off time and then re-transmits the data till later time slot

Throughput: $S = G \cdot e^{-2G}$ $\rightarrow$ throughput for successful transⁿ
 $V_t = T_f$ $G \rightarrow$ no. of transmissions

This means a frame is only at risk of collision during one time slot not before + after transmission like in Pure ALOHA

$\rightarrow$ max throughput: $\frac{dS}{dG} = 0 \Rightarrow G = 1 \Rightarrow S = 37\%$

Q) A n/w administrator is evaluating the performance of PURE ALOHA and slotted ALOHA protocols for a Commⁿ system. The N/w exp^{er} an avg of 1.2 transmissⁿ attempts per frame time. and the transmissⁿ time of each frame is 10 ms. (i) calculate throughput for both cases (ii) compare their efficiency & determine the % improvement in throughput of slotted ALOHA over PURE ALOHA

(i) $G = 1.2$, pure $\Rightarrow S_p = 1.2 \cdot e^{-2.4} = 0.1088 \Rightarrow 10.88\%$
slotted $\Rightarrow S_s = 1.2^{-1.2} = 0.3614 \Rightarrow 36\%$

(ii) % improvement = $\frac{S_{\text{slotted}} - S_{\text{pure}}}{S_{\text{pure}}} \times 100 = 232.2\%$

CSMA: a device checks whether the communication channel is already used before sending data or not
MA $\rightarrow$ Multiple access, it means that multiple devices share same communication medium

CSMA: Is a protocol where multiple devices share a common channel and each device checks the channel before transmission to make sure it is free

CSMA/CD

Feature:

1) Basic Approach

→ It detects collision after it happens

2) Working principle

→ Device checks the channel, sends the data and keeps monitoring, if a collision occurs, it stops & retransmits data

3) Collision Handling

→ Collision is detected & transmission is stopped immediately

4) N/w type

→ Used in wired N/w

5) Delay

→ Delay happens due to retransmission after collision

Controlled Access Protocol

→ In this multiple devices share a

to avoid collisions. Since multiple devices need to transmit data over the same communication channel, therefore, a controlled mechanism is required

→ Instead of ^{allowing} all devices to transmit freely the protocol ensures that only 1 device transmits at a time

→ In this approach device is not allowed to send data directly

→ It must obtain permission to access the channel and

→ The system that determines which device will transmit and allocates the transmission opportunity acc"

→ Once the permission is granted, the selected device transmits data while other devices wait for their turn

→ After the transmission is completed, control is transferred to next device. This control can be implemented using diff" methods such as i) polling (where a central controller selects devices 1 by one)

CSMA/CA

→ Tries to avoid collision before it happens

→ Device checks the channel waits for some time & then sends data after assuring it is safe

→ Collision is minimized by introducing waiting time before transmission

→ Used in wireless N/w

→ Delay occurs because of waiting time

ii) Token only

iii) Reservation

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→ IEEE 802.5 -

Ring topolog

ii) token Passing (where token circulates among devices & only the device holding the token can transmit data)

iii) Reservation (where time slots are allocated in advance)
This process continues in an orderly manner ensuring that all devices get a fair opportunity to transmit data

Channelization based Protocols:

→ The available BW of the commⁿ channel is divided into multiple smaller channels using different techniques

→ FDMA, TDMA & CDMA

→ Each device is allocated the dedicated portion of channel either in terms of frequency, time, code.

→ This allocation allows multiple devices to transmit data simultaneously without causing collision. As a result channelization ensures efficient utilization of BW &

provides collision free communication. Ex: FDMA, TDMA, CDMA

→ FDMA: channel is divided into diffⁿ freq^s bands

TDMA: — " — " — " — " time slots

CDMA: — " — " — " — " specific code for each device

IEEE standards:

→ IEEE stands for Institute of Electrical & Electronics Engineering

→ It is International organization that develops for standardization & Networking

→ IEEE 802: These are popular bcz they provide a collection of networking standards that defines how commⁿ takes place at the physical & Datalink layer

→ These standards are used in technologies like Internet and Ethernet

→ IEEE 802.1 → specifies → for N/w management, bridging & Interconnectⁿ b/w LAN's

→ IEEE 802.2 → Defines LLC sublayer responsible for error control

→ IEEE 802.3 → This specifies standard for wired LAN commⁿ using CMA/CD protocol. (Ethernet)

→ IEEE 802.5 → Defines token passing protocol for commⁿ in Ring topology.

- IEEE 802.11 : Specifies for wireless LAN communication (wifi related) using CSMA/CA commonly used in mobile.
- IEEE 802.15 → Bluetooth: Used for short range communication between devices
- IEEE 802.16 → Used for wireless metropolitan area networks (WMAN) covers large areas

Token Ring: is a Medium Access protocol used in LAN

- In this method all devices are connected in a logical ring topology and data transmission takes place in an organized & controlled manner using a special signal called as "Token"
- Token: A token is a small control frame that continuously circulates around the N/W. Usually its size is 3B
- Token provides permission to transmit data through the N/W only the device or station that holds the token is allowed to send data & all other devices must wait for their turn
- As the token circulates from one device to another, each device gets a fair chance to transmit data when it receives token.

06/04/2026

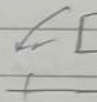
Token Passing Process:-

- In a Token Ring network, a small control frame (Token) keeps moving continuously from one station to other in a Ring.
- A station that wants to send data cannot transmit immediately, it must first wait for the token, when the token arrives, the station captures it and sends its data frame into the N/W. The data then travels around the Ring in one direction until it reaches the Destination station.
- The Destination copies data & sends acknowledgement. After completing the transmission the DF returns to sender which confirms that the data has been successfully delivered

⇒ The sender so that the
→ In this way data and

Token Bus topology is

- The network but the to a fixed a which station
- A station it holds the station



virtual ring

Token Bus

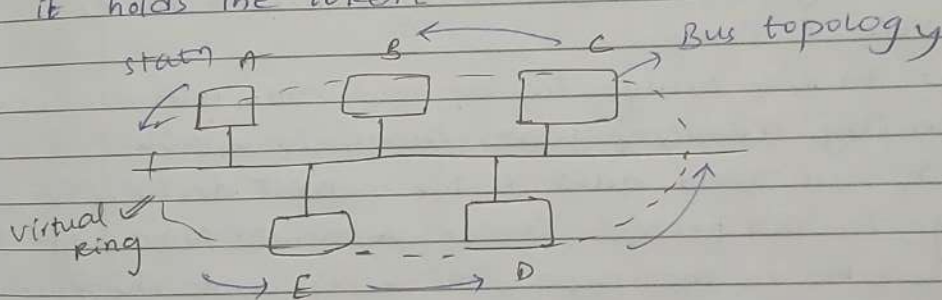
- Uses Bus topology physical
- all stations to a common
- Token is a predefined

* Bridge: N/W form a large OSI model.

⇒ The sender then releases the token back into the ring, so that the next station can use it. In this way each station gets a fair chance to transmit data and collisions are completely avoided.

Token Bus: In a token bus network the physical topology is a bus while the commⁿ follows

- The network uses coaxial cables for physical commⁿ but the token is passed among the stations in a fixed and sequential order. Each station knows which station comes before it & which comes after it.
- A station can only send or transmit data only when it holds the token.



Token Bus

- Uses Bus topology for physical connection
- all stations are connected to a common bus cable
- Token is passed in a predefined sequence

Token Ring

- Uses Ring topology
- Each station is connected to its neighbouring station
- Token is passed physically around the ring

* Bridge:

N/w device that connects multiple sub ^{single} n/w to form a large size n/w. It operates as DLL of OSI model.

→ A bridge ~~has~~ helps in reducing unnecessary n/w traffic by forwarding data, only to appropriate segments instead of sending it to all connected device. A Bridge plays an impⁿ role in n/w commⁿ by filtering & forwarding data frames based on their MAC address

→ Functions of Bridge:

- ① frame filtering: In this Bridge checks the destination MAC address of each frame & decides whether to forward or drop it
- ② forwarding: In this the Bridge sends data only to correct n/w segment where that destination device is available in table
- ③ Learning: In this Bridge learns the MAC Address from incoming frames & stores them in a table so that it can send data to the correct device in future
- ④ Flooding: In this process, if the destination MAC addⁿ is not available in the table, the bridge sends the frame to all connected N/w segment except the source port or device

Types of Bridges:

- ① Transparent Bridge
- ② Source Routing Bridge
- ③ Translational Bridge

① Transparent Bridge: This is the most commonly used type of Bridge in Ethernet N/w. It works in such a way that the connected devices are not aware of its presence

→ so that Bridge automatically learns the MAC addresses and forwards frames to correct destination
→ Used in LANs (lan abns)

② Source path

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- ② Source Routing Bridge: In this type of Bridge the complete path for data transmission is decided by sender itself.
- The sender includes the routing info in frame & the bridge simply follows that path to ~~send~~ ^{forward} data.
 - Used in 'Token Ring N/w'.
- ③ Translational Bridge: This type of Bridge is used to connect two diff^t types of N/w's that used diff^t topologies & protocols.
- It converts data format so that commⁿ can happen easily.

• NETWORK LAYER.

- 3rd Layer of OSI model
- Responsible for delivering data from source device to destination device across multiple N/w's.
- Deals with logical Address (IP addⁿ) and routing process.
- Logical Address helps in uniquely identifying each device in N/w, while routing is the process of finding the best possible path for data to travel from the source to destination.
- Network layer ensures that data reached correct destination efficiently even when multiple N/w's are involved.

Network layer

Data Link layer

→ error control & flow control

Physical layer

→ transmissⁿ of raw bits

Internetworking:

Process of connecting 2 or more Independent networks so that they can commⁿ with each other & work together as a single unified commⁿ system.

Key funⁿ of Internetworking:

- 1) Addressing: It assigns a unique ip address to each device so that it can be identified in the N/w
- 2) Routing: It determines the best path for data to travel from source to destination
- 3) Packet Forwarding: Once the path is decided the data is divided into small units called packets. These packets are then forwarded from one router to another untill they reach the final restination so basically, packets forwarding moves packer from one N/w to another based on routing info"
- 4) Protocol Translation: It allows commⁿ b/w diffⁿ type of N/w using diffⁿ protocols

Components of Internetworking:

- 1) Router: A router connects multiple N/w's and is responsible for deciding the path of data using Routing algorithms

Eg: BF, Linkstate algo, Distance vector algo, Dijkstra's

- 2) Switch: Mainly used to connect devices within the same N/w. Handles commⁿ within N types of

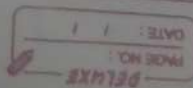
- 3) Gateway: Acts as a translator b/w diffⁿ N/w and or protocols. It enables commⁿ b/w N/w's that may use different technologies (standards)

** Protocols Involved in Internetworking Process

→ several impⁿ protocols are used in Intern/wing procw to ensure proper commⁿ b/w diffⁿ N/w's

- ① Internet Protocol (IP): IP is most fundamental protocol, it provides logical addressing and is responsible for delivering packets across N/w

→ IP is classified into 2 types: IPV₄ & IPV₆
IPV₄: most widely protocol at NW layer, it is responsible for identifying devices & delivering data from src → destination over the Internet



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class
class E

→ IPv₄ address is 32 bit logical address which uniquely identifies a device in NW. It is usually written in a dotted decimal form → 192.168.1.1

where address is divided into four parts, which is known as Octet. And each octet consists of 8 bits

→ And each octet is a number ranging from (0 → 255)

Remember IPv₄ ^{Provides} connectionless service

→ In IPv₄, data is sent in the form of packets i.e., called "Datagram" and each datagram is divided into 2 main parts one is Header & other is Payload (data)

→ Header contains impⁿ infoⁿ src address, destⁿ addⁿ and routing infoⁿ

15/01/26

IP $\left\{ \begin{array}{l} \text{IPv}_4 : 2^{32} \approx 4.3 \text{ billions unique addr} \\ \text{IPv}_6 \end{array} \right.$

IPv₆ :

→ IPv₆ is next generation of IP, it uses a 128 bit address which provides a large number of unique addresses

→ IPv₆ addrⁿ are written in hexadecimal format separated by columns. Ex:

Ex: 2001 : 0db8 : 85a3 : 0000 : 0000 : 8a2e : 0370 : 73348

→ It contains 8 groups, each grp is made of 4 Hexa-decimal digits, each hexadecimal digit contains 4 bits, so each grp in IPv₆ contain 16 bits

→ IPv₆ supports around 2^{128} unique addresses

#

Classful Addressing: is a method of IP addressing that is used only in IPv₄. In this method the IP addr is divided into 5 classes that is Class A, B, C, D, E

→ IP addr is divided into 5 cls based on the value of first octet.

Class A → First octet of class A starts with 0

Class B → " " B " " 10

Class C → " " C " " 110

Class D → " " D " " 1110

Class E → " " E " " 11110

Classwise	Binary Range	Decimal Range
Class A :	00000000 to 00000000 11111111	(0, 127)
Class B :	10000000 to 10 111111	(128, 191)
Class C :	110 00000 to 110 11111	(192, 223)
Class D :	111 0 0000 to 111 0 1111	(224, 239)
Class E :	1111 0000 to 1111 1111	(240, 255)

- Class A is used for large N/w's where a Very large no. of devices are required or connected
 - Class B is used for medium size N/w's
 - Class C " " Small " N/w's
 - Class D is used for Multicast Comm"
 - " E " " Research & Experimental purposes
- Ex: IP add :- (150). 20. 10. 5 → belongs to which class?
 ↳ belong to cls B ⇒ Class B

- IP addr consists of 2 main parts :
 - Network ID
 - Host ID
- Network ID identifies the network to which a device belongs. All the devices within the same N/w have same network ID
- Host ID identifies a specific device within that N/w & remember each device in same N/w ^{must} have a unique Host ID
- In class A, the 1st octet represent Network ID & the remaining 3 octets represent Host ID
- In class B, first two octets represent N/w ID & the rem" 2 rep" Host ID
- In class C, first three " " " " & the rem 1 (last octet) repr" Host ID
- In class D & class E there is no concept of N/w ID and Host ID

Routing

- is a process follow to
- During this contains in
- RA's have in forwarding
- RA is class
 - Non - A
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- Non - Adaptive is fixed an
- Network c
- The routes to all the
- Ex: Flooding
- Adaptive path change path dynamic
- Ex: Dijkstra

Dijkstra's

- Used to find all other
- In CN, i
- link state find the
- And this

Input:



Routing Algorithm: (RA)

- is a process used to determine the path that packet follow to reach the destination
- During this process, routing table is generated which contains info about different routes in the N/w
- RA's have in selecting the most efficient path for forwarding data packets from the source → destination
- RA is classified into 2 categories

i) Non-Adaptive (static) Routing

ii) Adaptive (Dynamic) Routing

⇒ Non-Adaptive Routing: In this type, the routing path is fixed and does not change according to N/w condition

→ network conⁿ → BW availability, Traffic, link failure

→ The routes are decided in advance & remain same to all the processes. But for small N/w's

Ex: Flooding Algo, Random Walk

⇒ Adaptive Routing Algo: In this type the routing path changes ^{based} on N/w conditions. It selects the best path dynamically. Generally used in large N/w's

Ex: ~~Dijkstra's~~ Dijkstra's, BF algo

Dijkstra's Algorithm

⇒ Used to find shortest path from given source node to all other source nodes in a N/w

→ In CN, it is mainly used for Routing algo in link state routing methods, where each router finds the best path

→ And this algo works only when link costs are +ve

Input:

* Dijkstra's Algo (DOM) ^{edges}

Input : $\rightarrow$ Network graph $G(V, E)$
 $\left\{ \begin{array}{l} \rightarrow \text{source node } s \quad \left\{ \begin{array}{l} \text{set of vertices} \end{array} \right. \\ \rightarrow \text{link cost } c(i, j) \end{array} \right.$

Output : $\rightarrow$ shortest distance from source to all ~~nodes~~ ^{nodes}
 $\rightarrow$ shortest path tree

Steps :

1) Initialization $\Rightarrow$ set distance of source node to 0, $D(s) = 0$
 $\rightarrow$ set distance of other nodes to ∞ ; $D(v) = \infty \quad \forall v \neq s$
 $\rightarrow$ create a ~~set~~ ^{set} of visited nodes N' , $N' = \phi$

Step 2: Update distance of neighbors
for each neighbour v of source s
 $D(v) = c(s, v)$

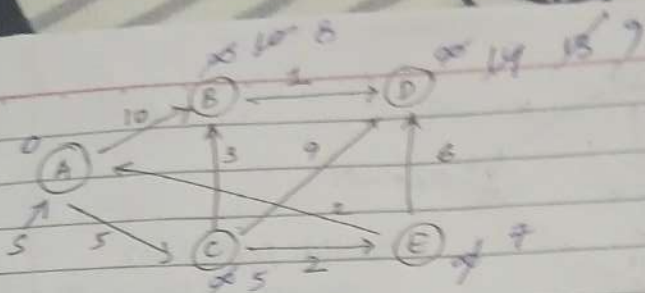
Step 3: Select Minimum distance Node
choose a node w such that ' $w \notin N'$ ' and it has
the minimum distance value

$D(w) \Rightarrow \min$
 $\rightarrow$ Add it to visited node N'
 $N' = N' \cup \{w\}$

Step 4: Update distances (or) Relaxation
for each unvisited neighbour v of node w
check if $D(w) + c(w, v) < D(v)$
update $D(v) = D(w) + c(w, v)$

Step 5: Repeat step 3 & 4 until all nodes are visited

Step 6: $D(v)$ gives shortest distance from source



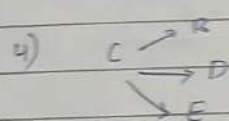
1) Node | Distance

A	0
B	∞
C	5
D	∞
E	∞

2) $N' = \{C\}$

$N' = \emptyset$

C STEP 2



$$D(B) + c(C, B) = \infty + 3 = \infty \quad D(B) = 8$$

$$D(C) + c(C, D) = 5 + 9 = 14 \Rightarrow D(D) = 14$$

$$D(C) + c(C, E) = 5 + 2 = 7 \Rightarrow D(E) = 7$$

$$D(E) = 7 \Rightarrow \min$$

$$N' = N' \cup \{E\} = \{C, E\}$$

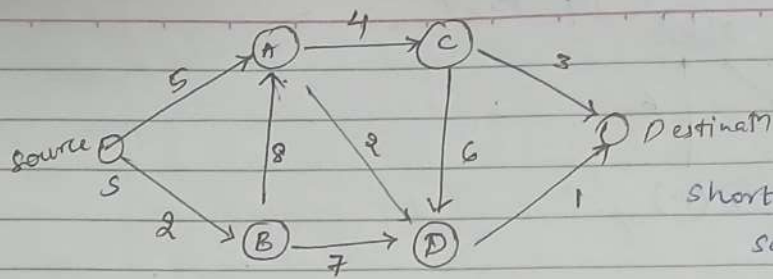
$$E \rightarrow D \Rightarrow D(E) + c(E, D) = 7 + 6 = 13 < D(D) \Rightarrow D(D) = 13$$

$$D(B) = 8 \Rightarrow \min, \quad B \rightarrow D: D(B) + c(B, D) = 8 + 1 = 9 < D(D) \Rightarrow$$

$$D(D) = 9$$

6) node | Distance

A	0
B	8
C	5
D	9
E	7



Shortest path from
source $\rightarrow$ Destination?

Node Distance

S	0	0✓	0✓
A	∞ 5	(5)	5✓
B	∞ (2)	2✓	2✓
C	∞ ∞	∞	9
D	∞ ∞	9	(7)✓
Dest ⁿ	∞	∞	∞ (8)✓

$$i) N' = \{S\}$$

$$ii) N' = \{S, B\}$$

$$iii) N' = \{S, B, A, D, \text{Dest}^n, C\}$$

$$D(\text{Dest}^n) = 8$$

Bell

$\rightarrow$ It is other

$\rightarrow$ If No this

{ if update

step 1

(A, 1)

step 2

$\rightarrow$ step 3

itera

(A)

(A)

(A)

*)

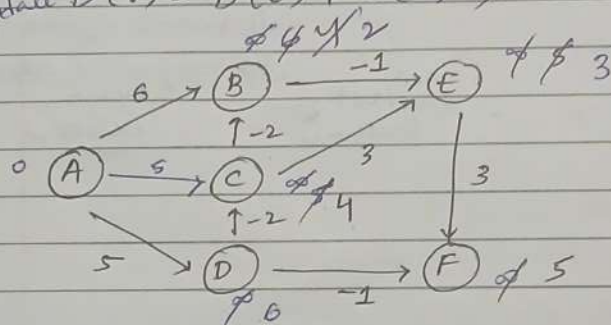
Bellmann ford : (Distance vector routing algo)

→ It is used to find shortest path from source router to other routers

→ If N/w graph contains both +ve & -ve edge weight this algorithm can find shortest path

{ if $D(u) + c(u,v) < D(v)$
update $D(v) = D(u) + c(u,v)$

→ done # $(N-1)$ times ^{# vertices}



step 1: write all edges of the graphs

$(A,B), (A,C), (A,D), (B,E), (C,B), (C,E), (D,C), (D,F), (E,F)$

step 2: same as Dijkstra

→ step 3: no. of iterations = $N-1 = 6-1 = 5$

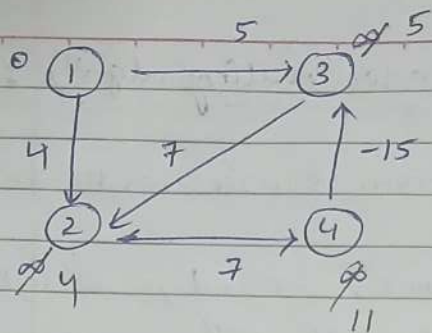
iteration - 1:

$(A \rightarrow B) \Rightarrow 0 + 6 < \infty \Rightarrow D(B) = 6$

$(A \rightarrow C) \Rightarrow 0 + 5 < \infty \Rightarrow D(C) = 5$

$(A \rightarrow D) \Rightarrow 0 + 5 < \infty \Rightarrow D(D) = 5$ by all edges

*)



$$n = 4 - 1 = 3$$

Edges:

(1, 2) (1, 3) (2, 4) (3, 2)
(4, 3)

no. of iterations should = $n - 1 = 4 - 1 = 3$

① ② ③ ④ ← do one more time

if values again change

No shortest
Distance

↳ negative Edge
cycle
exists